

**TRIPMATE:** A Mobile Travel Companion for  
motorists

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## **Introduction**

In today modern world, technology plays a vital role in almost every industry and in peoples daily lives. Among these industries, the transportation and personal mobility sector stand out as a critical area for technological innovation, as it directly impacts the safety, efficiency and convenience of daily commutes and long-distance travel. The demand for a dedicated mobile application like Tripmate arises in this technological age where immediate access to information is essential.

At present, many drivers and vehicle owners in the municipality of Buenavista, Bohol continue to face significant challenges in locating reliable automotive services. When experiencing vehicle troubles such as flat tires or engine problems, residents and visitors often rely on inefficient methods like word-of-mouth recommendations or time-consuming manual searches for vulcanizing shops and gas stations. This traditional approach frequently leads to extended waiting times, difficulty in finding trustworthy service providers and potential safety risks when vehicles break down in inconvenient or hazardous locations.

To respond to these challenges, the proponents proposed a Tripmate Mobile Application a comprehensive digital solution that will serve the entire municipality of Buenavista, Bohol. This innovative platform seeks to bridge the gap between vehicle owners and automotive services providers by creating an integrated network of gas stations and vulcanizing shops thereby enhancing road safety, improving service accessibility and supporting local economic growth.

## Purpose and Description

The project “TRIPMATE : A Mobile Travel Companion for motorists” was developed in order to provide motorists with a reliable and user-friendly mobile application that integrates essential travel features such as location-based service finders, emergency assistance and trip management tools. It aims to make traveling more convenient, efficient and safe for all types of motorists.

The System has the following capabilities:

- Local nearby fuel stations through real-time GPS and map integration, helping motorists easily find gas stations during their travels.
- File sharing and media streaming, which allows users to share important travel-related files such as maps, documents, or even photos and videos with fellow travelers while on the road.
- Real-time emergency vehicle support, which enables motorists to request help or notify emergency services quickly in case of accidents or roadside problems.
- Offer navigation and route planning to help users reach their destinations efficiently by providing the best routes and estimated travel times.
- Provide information on nearby repair shops and spare parts stores, ensuring motorists can easily locate assistance for vehicle issues.
- Save favorite destinations and display travel history for easy reference on future trips.
- Send alerts and notification about important updates, such as traffic conditions, nearby services, or safety reminders.

Once the proposed TRIPMATE is implemented, it will give significance to the following:

- Motorists – The system will make traveling easier and more convenient by providing all-in-one assistance through the app. It helps drivers save time, feels safer on the road, and be more confident knowing that help and information are just one tap away.
- Reasearchers – This study can serve as reference for those who wish to explore mobile travel applications, providing ideas on how technology can be used to promote safer and smarter driving experiences.
- Future Reasearchers – The project will serve as a guide and inspiration for future studies related to travel assistance, navigation and safety systems. It can help them improve or create new innovations that further enhance travel convenience and security.
- Community and Road Users – The project will contibute to safer roads and more organized travel experiences by providing motorists with a tool that promotes awareness, preparedness and quick access to help in case of emergencies.

## Objectives of the Study

This study aims to design, develop and evaluate TRIPMATE: A Mobile Travel Companion for Motorists, a mobile application that provides motorists with an all-in-one platform for safe, convenient and efficient travel.

Specifically, this study aims to achieve the following objectives:

- To design and develop the TRIPMATE mobile application that integrates the following major features:
  - Real-Time Fuel and Service Finder, which includes:
    - a). Fuel station locator with GPS and map integration.
    - b). Vulcanizing and repair shop locator.
    - c). Geolocation Tracking.
    - d). Emergency assistance and referral support.
  - Trip Management and Navigation Tools, which include:
    - a). Route planning and navigation with estimated travel time.
    - b). Saving favorite destinations and travel history.
    - c). File sharing and media streaming for travel-related content.
2. To implement a user-friendly and reliable mobile platform capable of providing real time data.
  3. To evaluate the overall functionality and performance of the TRIPMATE application in terms of:
    - a). Accuracy and responsiveness of location-based features.
    - b). Accuracy of Map Integration and Geolocation Tracking.
    - c). Efficiency of navigation and emergency assistance tools.

## Features:

### 1. Finder for Fuel Stations

- Displays stations using the GPS on your phone, along with information such as:
  - a). Types and costs of fuel.
  - b). Hours of operation.
- Present wait/queue times.
- Amenities that are available (convenience shop, bathrooms, air pump).

### 2. Locator for Spare Parts

- A search engine to locate particular auto parts at local stores or vendors.
- When users search for an item (for example, “Honda XRM 2015 brake pads”), the app displays the price and location of the vulcanizing or auto parts retailers that carry it.

### 3. Finder of Repair Services

- A booking and directory system for auto repair services and vulcanizing establishments. Users are able to:
  - a). View local stores that have been rated and reviewed.
  - b). Examine the services provided (brake, alignment and tire repair).
- Schedule a meeting or verify “walk-in” wait times.

### 4. Roadside Support

- When your automobile breaks down, you can use the emergency option to summon rescue. Users can ask for assistance for:
  - a). Change flat tires
  - b). Towing Service
  - c). Jump starters in batteries
  - d). Fuel delivery in an emergency

### 5. Tracker for Vehicle Maintenance after users enter their car’s information, the app:

- Sends out reminders for tire rotations, oil changes, etc.
- Records previous repairs and servicing. It keeps track of mileage and fuel efficiency.

### 6. Making Travel Plans

- When traveling a long distance, the app automatically recommends the best places to stop for:
  - a). Refueling

- b). Rest Periods
- c). Check-ups of the vehicle (e.g., tire pressure before to a mountain drive).

#### 7. Community Discussion Board

- Automobile enthusiasts and drivers can:
  - a). Request recommendations from mechanics.
  - b). Discuss your thoughts and experiences with particular stores.
  - c). Talk about typical auto issues and fixes.

#### 8. Alerts in Real Time

- Push alerts for significant changes. The app has the ability to notify users of:
  - a). Road closures or traffic bottlenecks in your area.
  - b). Short-term discounts on auto repair or fuel.
  - c). Notifications or recalls for your particular car model.

#### 9. Connectivity to Maps

- Navigation to a repair shop or fueling station would be launched instantly if you tapped on it. A turn-by-turn route would be automatically generated via travel planning.